



**Tribhuvan University**

**Faculty of Humanities and Social Science**

**AN INTERNSHIP REPORT**

**ON**

**DEVELOPMENT OF MODULAR UI INTERFACES FOR BUSINESS SOLUTIONS**

**At**

**Sarbatra Inc**

**Submitted to**

**Department of Computer Application**

**VS International College**

*In partial fulfillment of the requirements for the Bachelors in Computer Application*

**Submitted by**

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## Table of Contents

|                                                            |     |
|------------------------------------------------------------|-----|
| LIST OF FIGURES .....                                      | iii |
| CHAPTER 1: INTRODUCTION .....                              | 1   |
| 1.1 Introduction.....                                      | 1   |
| 1.2 Problem Statement.....                                 | 1   |
| 1.3 Objectives .....                                       | 2   |
| 1.4 Scope and Limitation .....                             | 2   |
| 1.4.1 Scope of the System .....                            | 2   |
| 1.4.2 Limitations of the System .....                      | 2   |
| 1.5 Development Methodology.....                           | 3   |
| 1.6 Report Organization .....                              | 4   |
| CHAPTER 2: Introduction to Organization.....               | 5   |
| 1.1 Introduction.....                                      | 5   |
| 1.2 Vision .....                                           | 5   |
| 1.3 Mission.....                                           | 5   |
| CHAPTER 3: DESCRIPTION OF INTERNSHIP WORK / PROJECT.....   | 6   |
| 3.1 System Overview.....                                   | 6   |
| 3.2 Technical Stack.....                                   | 7   |
| 3.3 Key Design Principles .....                            | 7   |
| 3.4 System Workflow .....                                  | 7   |
| 3.5 Gantt Chart .....                                      | 8   |
| CHAPTER 4: INTERNSHIP ACTIVITIES .....                     | 10  |
| 4.1 Roles And Responsibilities: .....                      | 10  |
| 4.2 Weekly Log.....                                        | 10  |
| 4.3 Description of Projects .....                          | 10  |
| CHAPTER 5: EXPECTED OUTCOME OF INTERNSHIP ACTIVITIES ..... | 11  |
| CHAPTER 6: CONCLUSION.....                                 | 13  |
| REFERENCES.....                                            | 14  |
| APPENDICES.....                                            | 15  |

**LIST OF FIGURES**

*Figure 1 :Gantt Chart* .....9

# **CHAPTER 1: INTRODUCTION**

## **1.1 Introduction**

An internship serves as a vital professional learning experience, offering meaningful and practical work that aligns with a student's academic background and career aspirations. It provides a unique opportunity to apply theoretical knowledge to complex, real-world problems while developing essential industry-specific skills.

This internship report focuses on the Development of Modular UI Interfaces for Business Solutions. In the modern digital landscape, a professional web presence requires more than static pages; it demands a scalable, component-based architecture that enhances user engagement and simplifies maintenance. My work at Sarbatra Inc. involved designing and implementing a system where key user interface elements—such as the Nav-bar, Product Showcase, and Wishlist—function as independent, reusable modules.

The project emphasizes the creation of a secure, dynamic, and highly responsive web application using HTML, Tailwind CSS, and JavaScript. By utilizing a custom component-loading mechanism, the system ensures that different business modules are injected dynamically into the main application structure, optimizing performance and code organization. This project has provided practical knowledge of Asynchronous JavaScript, Utility-first CSS styling, and Responsive Layout Design, all of which are fundamental skills for a successful career in modern web and software development.

## **1.2 Problem Statement**

In the modern world, businesses need websites that look professional and work perfectly on all devices. However, many websites are built in a way that makes them very hard to fix or update. This creates several problems:

- **Maintenance Complexity:** Updating a single element, such as a site-wide navigation bar or a footer, requires manual changes across multiple pages, increasing the risk of inconsistency and errors.
- **Screen Size Issues:** Many older websites do not look good on mobile phones or tablets, which makes it hard for customers to use them.
- **Poor Scalability:** As business requirements grow, adding new features or sections becomes difficult and time-consuming because the existing code structure is rigid and not designed for expansion.

### 1.3 Objectives

- To design and develop a responsive business interface.
- To prepare for modern frameworks.
- To gain professional experience.

### 1.4 Scope and Limitation

#### 1.4.1 Scope of the System

- The scope of this internship project defines the boundaries of the modular UI system.  
The project includes:
- Development of a modular web-based interface
- Modular section integration
- Dynamic Component Loading
- Utility-First Styling:
- Frontend-to-Framework Transition

The project is designed to serve as a foundational system that demonstrates modern, scalable frontend development practices suitable for business solutions

#### 1.4.2 Limitations of the System

The proposed system has certain limitations due to its academic nature and time constraints:

- The project is developed within a limited internship duration
- Only a single admin user is supported
- Advanced security features such as two-factor authentication are not included
- No integration with third-party APIs or cloud services
- The system is intended for learning and demonstration purposes
- Scalability for high-traffic commercial deployment is not considered

## **1.5 Development Methodology**

The development of this project followed the **Agile Methodology**. This approach focuses on building the software in small parts and improving them through constant testing and feedback.

### **Requirement Analysis**

Identifying the essential UI components needed for the business solution, such as the Nav-bar, Product Grid, and Wishlist.

### **Design Phase**

This phase involves designing overall system architecture. Creating the visual layout and selecting the right color schemes and typography using Tailwind CSS to ensure a professional look.

### **Development**

Building each module one by one (e.g., Week 1: Header, Week 2: Products). Each module was developed as a separate HTML file for better organization.

### **Testing & Debugging**

Checking the website on different screen sizes (mobile vs. desktop) and fixing any layout or JavaScript errors found in the browser.

### **Review**

Presenting the completed modules to the supervisor at Sarbatra Inc. and making changes based on their feedback.

## **1.6 Report Organization**

This internship proposal defense report is organized in a structured manner to ensure clarity and logical flow:

- **Chapter 1** presents the introduction, problem statement, objectives, scope and limitations, development methodology, and report organization.
- **Chapter 2** describes the introduction of the organization, vision, mission.
- **Chapter 3** describes the expected internship work and provides an overview of the proposed system.
- **Chapter 3** explains system overview, and project scheduling through a Gantt chart.
- **Chapter 4** discusses the internship activities.
- **Chapter 5** discusses the expected outcomes of the internship.
- **Chapter 6** explains the conclusion of the report.

## **CHAPTER 2: Introduction to Organization**

### **1.1 Introduction**

Sarbatra Inc. is a premier IT solution provider based in Kathmandu, Nepal, dedicated to helping businesses navigate the complexities of the digital age. The company offers a comprehensive suite of services, including web and mobile application development, custom software engineering, and innovative digital marketing strategies such as SEO and social proofing. By blending creative design with cutting-edge technology—including their proprietary Sarbatra CMS and "Social Signal" tools—the firm empowers brands to optimize their operations, enhance their online authority, and achieve sustainable growth in a competitive global market.

### **1.2 Vision**

To be a global leader in innovative technology by delivering simple, high-quality, and user-friendly digital experiences to businesses of all sizes.

### **1.3 Mission**

To deliver practical, reliable, and accessible technology to everyone by building long-term relationships through transparent communication and exceptional results.



## CHAPTER 3: DESCRIPTION OF INTERNSHIP WORK / PROJECT

### 3.1 System Overview

The architecture follows a modular approach where the main structure is maintained in a central entry point, while specific business logic and UI elements are separated into individual files.

- **Main Entry Point (app.html):** Acts as a container with designated div placeholders for various system modules like the navigation bar, product grid, and checkout sections.
- **Dynamic Component Loader (component-loader.js):** The core engine of the system that fetches and injects HTML/JS components into the main page at runtime.
- **Modular Layers:** The UI is divided into six distinct functional blocks:
  - **Top-bar & Header:** Branding and notification layer.
  - **Nav-bar:** Global navigation control.
  - **Products & Wishlist:** Core business modules for data display and user interaction.
  - **Features & Footer:** Informational and closing sections.

### 3.2 Technical Stack

The system is built using modern, lightweight technologies to ensure high performance and scalability:

- **Frontend Foundations:** HTML5 and JavaScript are used for structure and dynamic behavior.
- **Styling:** Tailwind CSS provides a utility-first approach for rapid, responsive design across all devices.
- **Dynamic Logic:** Frontend Foundations: HTML5 and JavaScript are used for structure and dynamic behavior.

### 3.3 Key Design Principles

- **Modularity:** Each UI interface (e.g., the Wishlist or Products section) is an independent module, making the system easy to maintain and debug.
- **Scalability:** New business features can be added simply by creating a new component and adding a placeholder in the main file.
- **Responsiveness:** The use of Tailwind CSS ensures that the business solutions remain accessible and user-friendly across mobile, tablet, and desktop views.

### 3.4 System Workflow

1. **Request:** The user opens app.html.
2. **Initialization:** The component-loader.js script triggers automatically.
3. **Assembly:** The script fetches individual modules (Top-bar, Nav-bar, Products, etc.) and injects them into their respective id containers.
4. **Final Render:** The user sees a fully assembled, cohesive business interface.

### 3.5 Gantt Chart

The Gantt chart illustrates the actual timeline for completing the internship project at **Sarbatra Inc.** It outlines the various phases of the work, starting from the initial setup to the final development of modern web modules. The chart serves as a planning and tracking tool to ensure the timely completion of all project tasks within the 5-week period.

The timeline is divided into the following key phases:

- **Environment Setup & Orientation:** The first week was dedicated to preparing the development tools and understanding the company's professional workflow.
- **Core UI Development:** During the second and third weeks, the focus was on designing and building the six modular interfaces using **HTML and Tailwind CSS**.
- **React Module Conversion:** Starting from the third week, the project transitioned into moving these static parts into a **React framework** to make them more dynamic and professional.



*Figure 1 :Gantt Chart*

## CHAPTER 4: INTERNSHIP ACTIVITIES

### 4.1 Roles And Responsibilities:

As a Frontend Intern, I was responsible for designing and developing modular web components. My tasks included implementing responsive layouts with Tailwind CSS and using JavaScript to handle dynamic logic and component loading.

### 4.2 Weekly Log

- **Weeks 1:** Environment setup and orientation to company coding standards.
- **Weeks 2-3:** Developed six core UI components: Top-bar, Header, Nav-bar, Wishlist, Product Showcase, and Footer.
- **Weeks 4-Present:** Initiated the transition of static components into React modules.

### 4.3 Description of Projects

The project involved building a modular architecture in app.html that uses a central script to load diverse UI modules. This approach ensures that each section—from the Checkout to the Features area—is easy to debug and update individually.

## CHAPTER 5: EXPECTED OUTCOME OF INTERNSHIP ACTIVITIES

The completion of the internship activities at Sarbatra Inc. is expected to yield the following technical and professional results:

### 1. Technical Proficiency in Moder Frontend Stack

- **Modular Component Architecture:** Successful development of a scalable web structure where UI elements (Top-bar, Nav-bar, Products, etc.) are decoupled and reusable.
- **Responsive Design Mastery:** Ability to create "Mobile-First" interfaces using **Tailwind CSS**, ensuring the business solutions are accessible across all device resolutions.
- **Dynamic Data Rendering:** Proficiency in using **JavaScript** (and component-loader.js) to asynchronously fetch and display data without full page reloads.
- **Transition to React:** A solid foundation in React.js, converting static HTML components into dynamic, state-managed React components to meet high-end industry standards.

### 2. Professional and industrial Skills

- **Workflow Integration:** Familiarity with the Professional Software Development Life Cycle (SDLC) followed at Sarbatra Inc., including agile methodologies and daily stand-ups.
- **Tool Competency:** Effective use of industry-standard tools such as VS Code, Git/GitHub, Chrome DevTools, and project management software.
- **Optimization Techniques:** Understanding how to optimize code for faster loading speeds and better SEO performance, aligning with Sarbatra's "Social Signal" and digital marketing goals.

### 3. Problem Solving and Soft Skills Development

- **Debugging Skills:** Improved ability to identify and resolve cross-browser compatibility issues and JavaScript logic errors.

- **Time Management:** Meeting production deadlines for component delivery using time-tracking tools
- **Communication:** Enhanced ability to communicate technical ideas clearly with senior developers and team leads.

## **CHAPTER 6: CONCLUSION**

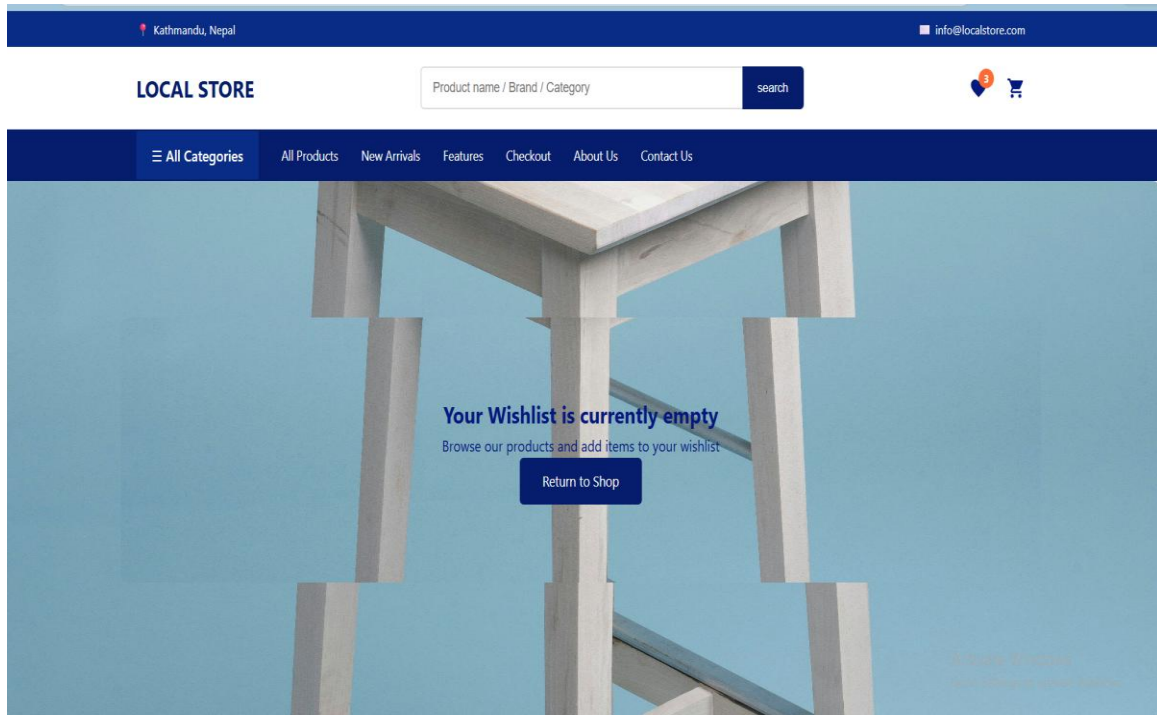
My internship at Sarbatra Inc. has been a transformative experience, bridging the gap between academic theory and industrial practice. By developing six modular UI interfaces, I learned the importance of clean code, scalability, and mobile-first design. This journey from using a simple `component-loader.js` to exploring React has provided me with a solid foundation in frontend engineering. I have not only improved my technical skills in Tailwind CSS and JavaScript but also developed professional soft skills like time management and team collaboration.



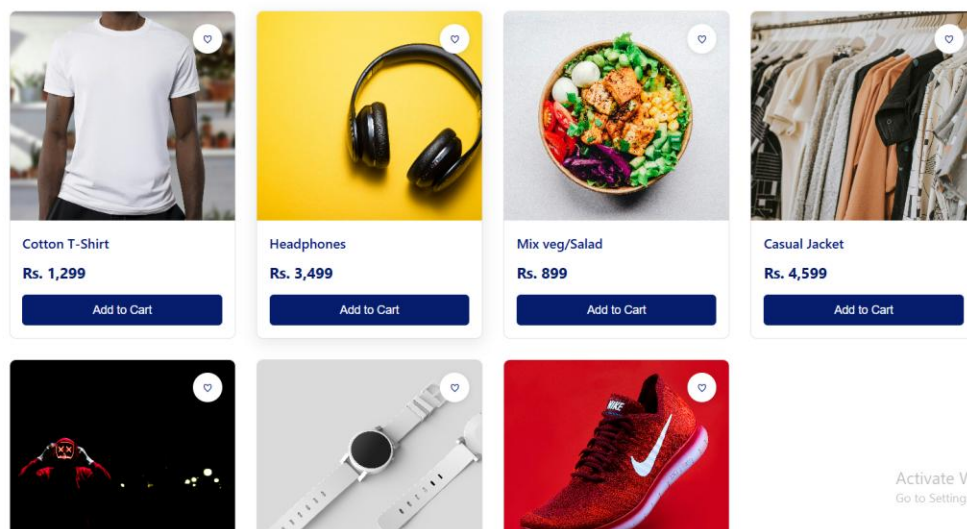
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## APPENDICES



### Featured Products



## Checkout

### Billing Details

First Name \*

Last Name \*

Company Name (Optional)

Country / Region \*

Street Address \*

Town / City \*

State \*

ZIP Code \*

### Your Order

Subtotal Rs. 0.00

Shipping Rs. 100.00

Tax Rs. 0.00

**Total Rs. 100.00**

Place Order

Activate Windows  
Go to Settings to activate Windows.



#### Support 24/7

Contact us anytime, anywhere



#### 100% Safety Payments

Your payments are secure



#### On-time Delivery

Fast and reliable shipping



#### No Minimum Orders

Order as much as you need

### LOCAL STORE

Your trusted local e-commerce partner for quality products. We deliver excellence in every order, ensuring customer satisfaction with every purchase.

f ig t in

### Quick Links

Shop  
Wishlist  
Checkout  
Contact info  
Privacy Policy  
Terms & Conditions  
FAQ

### Contact Info

- Kathmandu, Nepal
- +977 1-4444444
- +977 98xxxxxxx
- info@localstore.com
- Mon-Sat: 9AM - 8PM
- Sunday: 10AM - 6PM



### Online Payment

We accept secure online payments through:

eSewa

FonePay

Click Here to Pay

Activate Windows  
Go to Settings to activate Windows.